A-A-1543A
May 14, 1982
SUPERSEDING
A-A-1543
May 14, 1980

COMMERCIAL ITEM DESCRIPTION

SOAP, BORAX (POWDERED)

The General Services Administration has authorized the use of this commercial item description in preference to Federal Specification P-S-625.

This description covers a powdered hand cleaner product, consisting of soap and borax for use in washroom dispensers to remove heavy occupational oils such as oils and greases.

The product shall be of the following classes:

Class 1 - Plain
Class 2 - With Lanolin

Salient characteristics:

The product shall conform to ASTM D 3046 - powdered hand cleaner (Soap with Borax), except that paragraph 3 2 shall not apply.

Class 1. Class 1 shall comply with Type A of ASTM D 3046.

Class 2. Class 2 shall comply with Type B of ASTM D 3046.

The issue of ASTM D 3046 in effect on the date of the solicitation shall be used to determine compliance with this requirement.

Preservation, packaging, packing, labeling, and marking. Preservation, packaging, packing, labeling, and marking shall be as specified in the contract or order

ASTM D 3046 is available from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103

NOTE The test for lanolin shall be conducted in accordance with the following method

Percent lanolin. The percent lanolin is determined colorimetrically based on the cholesterol present in lanolin Cholesterol in chloroform solution in the presence of acetic anhydride and concentrated sulfuric acid produces a color which varies in intensity with the cholesterol concentration Total cholesterol in the sample is determined, corrected for the cholesterol present in the soap, and the percent lanolin in the sample calculated.

Apparatus. Any commercial colorimeter having an approximate spectral range of 640 to 700 nanometers may be used to determine the absorbance of the samples. The instrument shall be adjusted to zero using as a blank 25 ml of chloroform treated as specified in the color development section, omitting the lanolin

Standard calibration curve. A sample of the batch of the lanolin used by the manufacturer to make the product being tested shall be collected at the time of manufacture and used to make the standard calibration curve Place 0 2 g of this lanolin into a tared 50-ml beaker and weigh to the nearest milligram Dissolve the lanolin in chloroform, transfer quantitatively to a 200-ml volumetric flask and make up to the mark with chloroform Pipet exactly 3 0, 6.0, 9.0, 12 0 and 15.0 ml quantities of this lanolin solution into separate 50-ml glass-stoppered graduated cylinders. Add chloroform to each cylinder to yield 25 ml of solution in each and develop the color as specified below Prepare a standard curve by plotting colorimeter readings versus the milligrams of lanolin in the graduated cylinder.

Procedure. Make all determination in duplicate

Sample preparation Place 10 g each of the well-mixed product and the soap portion into separate tared soxhlet extraction thimbles and weigh to the nearest milligram Extract with petroleum ether in a soxhlet apparatus for 3 hours Evaporate the petroleum ether solution to dryness on a steam bath Dissolve the residue in chloroform, transfer quantitatively to a 250-ml volumetric flask and make up to the mark with chloroform. Pipet 10 ml of the chloroform solution into a 50-ml glass-stoppered cylinder and add chloroform to the 25 ml mark

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Color development. Cool the cylinder in an ice bath for 5 minutes. Pipet 10 ml of acetic anhydride into the cylinder and then add concentrated sulfuric acid dropwise from a 1-ml graduated pipet at a rate of 2 drops per minute for 7 minutes and 3 drops per minute for the next 3 minutes, stoppering the cylinder after each addition and mixing thoroughly by inverting the cylinder several times. Keep the cylinder in the ice bath between additions of acid. Remove the cylinder from the bath and store in a dark place for exactly minutes and then determine the color intensity in the colorimeter.

Calculations For every sample of the finished product and soap portion calculate the "equivalent percent lanolin" as follows:

"Equivalent percent lanolin" = 2.5 x (A/B)

A = mg of lanolin from the standard curve equivalent to the measured color intensity of the sample.

B = weight in grams of the sample

The percent lanolin in the finished product shall be calculated as follows:

Percent lanolin = C - DxE/100

where:

C = average "equivalent percent lanolin" in the finished product.

D = average "equivalent percent lanolin" in the soap portion

F = percent soap in the finished product.

MILITARY INTEREST:

Custodians

Army - GL
Air Force - 99

Review Activity

Army - MD

CIVIL AGENCY COORDINATING ACTIVITIES

GPO
GSA-FSS, PBO
USPS-POS
VA - OSS

PREPARING ACTIVITY

GSA-FSS

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